

Reclaiming Korea's Vanishing Coastline: Sustainable Coastal Protection Beyond Manmade Structures

Last summer, visitors to Gyeongpo Beach in Gangneung encountered an unexpected sight. The formerly expansive shoreline had significantly receded, with several sections suffering from severe erosion and restricted access. This phenomenon is not unique. More than 43 percent of the country's 360 surveyed coastal areas were classified as suffering "at risk" or "severe" coastal erosion. Despite this reality, South Korea continues to rely predominantly on artificial structures, such as seawalls, revetments, and breakwaters, as the primary method of coastal protection. However, this structure-centric approach is increasingly unsustainable. In an era characterized by rising sea levels and extreme weather events, it is imperative to transition toward nature-based coastal management strategies as a durable and resilient policy alternative.

Climate science has already issued a warning that these alternatives are needed. The Intergovernmental Panel on Climate Change projects continued acceleration in global sea-level rise this century, with worst-case end-of-century increases approaching or exceeding one meter, putting many existing coastal structures under growing strain.¹ Additional experts state that hard-engineering structures alone cannot prevent coastal erosion. While seawalls and breakwaters are capable of diminishing wave impacts temporarily, they also interfere with littoral drift, thereby accelerating erosion in neighboring regions. On certain beaches along Korea's eastern coastline, where beach widths decreased by almost 20 meters within five years following the installation of improperly designed or inadequately functioning coastal structures.²

Apart from engineering, nature-based solutions, such as restoring wetlands, expanding salt marshes, or reconstructing dune systems, offer effective wave attenuation while maintaining ecological integrity and decreasing long-term maintenance expenses. Nations such as the Netherlands, the United States, and Australia have demonstrated significant success with these approaches, attaining both diminished erosion and enhanced coastal resilience. For instance, the Netherlands' Sand Motor Project, a large-scale sand nourishment implemented in 2011, has effectively reclaimed eroded beaches by depositing large volumes of sand that are gradually redistributed by natural coastal processes.³ Similarly, coastal wetlands in the northeastern United States reduced storm surge impacts during Hurricane Sandy, preventing more than \$625 million in property damages and reducing flood losses by about 10% on average.⁴

In the short term, coastal structures may appear to be the most dependable barrier against rising sea levels and storm surges. However, the perceived immediacy of this protection can be deceptive.

¹ (NOAA 2023)

² (Shim and Kim 2024)

³ (Taal, et al. 2016)

⁴ (Resilience 2026)

These structures frequently induce increased erosion in adjacent regions, thereby elevating the overall risk. Furthermore, maintenance and repair expenses are expected to escalate significantly as sea levels continue to rise. For example, global coastal protection through dikes alone could require annual investment and maintenance costs of US\$ 12-71 billion in 2100. Moreover, the study indicates that future flood damages are far more sensitive to the protection strategies, underscoring the importance of long-term coastal adaptation strategies.⁵ Thus, the stability conferred by coastal structures may, in many respects, be a temporal illusion.

What Korea needs is not bigger or higher seawalls but smarter policy decisions. The national government should integrate nature-based coastal management into its long-term climate adaptation strategy, while employing human-made coastal structures more selectively and strategically where they are genuinely necessary. Local governments must shift away from short-term, construction-focused solutions and instead invest in restoring natural systems or focus on creating environments that support climate-resilient plant species, thereby strengthening the resilience of coastal ecosystems over time. Coastlines are more than tourist attractions: they are a frontline defense against climate-related disasters.

Across the world, it is increasingly common for nations to implement nature-based approaches to coastal management. Korea may consider these examples and draw lessons from these experiences. Climate change remains an unavoidable reality; however, the destiny of global coastlines is within our control. The decisions made today will influence whether coastlines disappear or are restored for future generations. Ultimately, effective coastal protection will depend not only on engineering solutions but also on forward-looking policy decisions.

⁵ (J. Hinkel 2014)

Bibliography

- J. Hinkel, D. Lincke, A.T. Vafeidis, M. Perrette, R.J. Nicholls, R.S.J. Tol, B. Marzeion, X. Fettweis, C. Ionescu, & A. Levermann. 2014. "Coastal flood damage and adaptation costs under 21st century sea-level rise." *National Academy of Sciences* 3292-3297.
- NOAA. 2023. *NOAA Climate.gov*. August 22. Accessed August 22, 2023. <https://www.climate.gov/news-features/understanding-climate/climate-change-global-sea-level>.
- Resilience, Coastal. 2026. *Coastal Resilience*. March 6. <https://coastalresilience.org/coastal-wetlands-provide-significant-flood-damage-reduction/>.
- Shim, Kyu-Tae, and Kyu-Han Kim. 2024. "A Study on the Effects of Morphological Changes Due to the Construction of Multiple Coastal Structures." *Journal of Marine Science and Engineering* 2174.
- Taal, M.D, M.A.M. Löffler, C.T.M. Vertegaal, J.W.M. Wijsman, L. Van der Valk, and P.K. Tonnon. 2016. *Development of the Sand Motor*. Deltares.